Lexical Functions to Improve Lexical Competence

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1. Introduction & Motivation

- **The Problem:** Large Language Models (LLMs) perform very well on general tasks, but their fine-grained understanding of semantic and lexical relations remains limited compared to human native speakers.
- **Proposed Solution:** Utilize the Meaning-Text Theory (MTT) and its **Lexical Functions (LFs)** to structurally inject lexical knowledge into models.
- **Objective:** Evaluate whether explicit fine-tuning on a structured dataset improves both specific lexical competence and general Natural Language Understanding (NLU).

2. Methodology

- **Dataset:** Data extracted from the *French Lexical Network (fr-LN)*, formatted in JSON (POS, senses, LFs, in-context examples). 80% train / 20% test split.
- **Evaluated Models:**
  - **Mistral-7B** (Modern, multilingual)
  - **Qwen2.5-7B** (Instruction-tuned, multilingual)
  - **FlauBERT** (French-specific, Masked Language Model)
- **Approach:** Parameter-Efficient Fine-Tuning (**LoRA**) to inject MTT constraints without prohibitive computational costs ($r = 16, \alpha = 32$).
- **Baselines:** Base *un-finetuned* models and models fine-tuned on unstructured *Wikipedia* text (to isolate the impact of MTT structure versus mere exposure to the French language).

3. COLE Benchmark

	FlauBERT Base	FlauBERT Wiki	FlauBERT LF	Qwen2.5 Base	Qwen2.5 Wiki	Qwen2.5 LF	Mistral Base	Mistral Wiki	Mistral LF
allocine	54.0	56.0	57.0	70.0	90.0	96.0	82.0	79.0	0.0
daccord	100.0	100.0	100.0	0.0	0.0	0.0	0.0	0.0	0.0
fracas	0.0	0.0	0.0	62.0	44.0	63.0	54.0	52.0	0.0
french_boolq	100.0	100.0	100.0	0.0	0.0	0.0	0.0	0.0	0.0
gqnli	44.4	44.4	55.6	40.7	33.3	37.0	22.2	33.3	0.0
lingnli	100.0	100.0	100.0	47.0	11.0	60.0	75.0	61.0	0.0
mms	100.0	100.0	100.0	0.0	0.0	0.0	0.0	0.0	0.0
mnli-nineeleven-fr-mt	100.0	100.0	100.0	0.0	0.0	0.0	0.0	0.0	0.0
multiblimp	53.0	53.0	53.0	89.0	90.0	84.0	65.0	63.0	21.0
paws_x	45.0	45.0	45.0	73.0	39.0	69.0	45.0	32.0	1.0
qfrblimp	59.0	59.0	59.0	81.0	78.0	77.0	60.0	61.0	41.0
qfrcola	70.0	70.0	69.0	78.0	61.0	77.0	72.0	71.0	0.0
rte3-french	25.0	27.0	32.0	61.0	23.0	53.0	37.0	43.0	0.0
sickfr	18.0	18.0	15.0	69.0	66.0	62.0	27.0	25.0	0.0
sts22	19.0	19.0	21.0	14.0	25.0	23.0	13.0	12.0	0.0
wino_x_lm	100.0	100.0	100.0	0.0	0.0	0.0	0.0	0.0	0.0
wino_x_mt	100.0	100.0	100.0	0.0	0.0	0.0	0.0	0.0	0.0
xnli	33.0	35.0	31.0	70.0	75.0	62.0	47.0	43.0	0.0
SCORE COLE (TOTAL)	62.2	62.6	63.2	41.9	35.3	42.4	33.3	32.0	3.5

Key Observations:

- **Evaluation Format Mismatch:** Encoders (FlauBERT) naturally excel at COLE's exact classification tasks, while decoders (Qwen/Mistral) often fail artificially (scoring 0.0%) due to strict expected-token constraints rather than true capability deficits.
- **Catastrophic Forgetting in Mistral:** Mistral LF's score collapsed to 3.5%, scoring 0.0% on almost all tasks, proving it overlearned the LF structure at the expense of general language modeling.
- **Minor General Gains:** Qwen LF showed slight NLU improvements over its base model, while FlauBERT's scores remained mostly stable across configurations.

4. LF Benchmark

	FlauBERT Base	FlauBERT Wiki	FlauBERT LF	Qwen2.5 Base	Qwen2.5 Wiki	Qwen2.5 LF	Mistral Base	Mistral Wiki	Mistral LF
A_0	2.7	8.2	3.9	12.9	6.2	67.2	58.6	60.9	25.0
A_1	5.6	12.5	9.7	16.9	16.1	34.7	9.7	7.7	7.3
A_2	9.1	9.1	9.1	6.8	6.8	34.1	18.2	31.8	13.6
Able_1	4.5	15.9	6.8	0.0	0.0	38.6	13.6	4.5	11.4
Able_2	6.2	6.2	7.8	4.7	1.6	57.8	37.5	40.6	18.8
Adv_0	2.6	2.2	2.6	12.1	2.2	76.3	38.8	56.0	22.4
Adv_1	25.0	14.3	10.7	10.7	0.0	60.7	46.4	46.4	14.3
Anti	4.4	10.3	10.3	5.9	4.4	58.8	34.6	47.8	24.3
AntiBon	11.1	13.9	12.5	0.0	2.8	12.5	11.1	11.1	0.0
AntiMagn	14.3	21.4	17.9	0.0	0.0	0.0	3.6	7.1	3.6
AntiVer	16.7	16.7	16.7	0.0	0.0	4.2	0.0	4.2	4.2
Bon	35.7	46.4	25.0	0.0	0.0	0.0	0.0	0.0	0.0
Contr	3.4	3.4	3.4	3.4	2.2	28.0	30.2	34.1	10.3
Conv_21	11.1	30.6	33.3	13.9	12.5	11.1	11.1	8.3	6.9
Fact_0	12.5	12.5	12.5	0.0	0.0	21.9	9.4	12.5	15.6
Fact_2	14.1	18.8	21.9	0.0	0.0	53.1	14.1	12.5	21.9
Gener	2.1	5.6	5.6	25.7	36.1	32.6	31.9	32.6	24.3
Hypo	10.0	32.5	10.0	10.0	5.0	7.5	2.5	2.5	5.0
Loc_in	50.0	50.0	50.0	20.8	29.2	54.2	87.5	79.2	37.5
Magn	3.3	7.6	8.7	3.3	2.2	4.3	6.5	5.4	1.1
Mero	8.3	8.3	14.6	6.2	4.2	18.8	29.2	31.2	8.3
Mult	9.1	9.1	9.1	9.1	4.5	15.9	31.8	29.5	22.7
Oper_1	4.3	3.4	3.4	0.4	1.3	15.9	8.2	8.6	4.7
Real_1	5.7	4.7	6.2	1.6	0.5	24.5	14.6	16.7	11.5
Redun	7.8	12.5	9.4	0.0	3.1	9.4	18.8	10.9	0.0
S_0	3.0	3.4	4.1	24.3	15.5	39.2	43.9	41.9	12.8
S_1	3.8	5.3	4.9	11.7	9.5	20.5	26.5	25.8	13.6
S_2	3.2	3.2	3.2	8.9	10.5	19.4	15.7	16.5	6.0
S_3	7.1	7.1	7.1	15.2	5.4	13.4	15.2	18.8	5.4
S_instr	13.9	13.9	19.4	22.2	11.1	22.2	33.3	30.6	22.2
S_loc	5.5	7.0	4.7	15.6	13.3	28.1	21.1	27.3	11.7
S_res	10.0	10.0	10.0	17.5	25.0	13.8	15.0	13.8	1.2
Sing	7.7	7.7	7.7	0.0	0.0	11.5	9.6	11.5	3.8
Syn	5.6	12.5	11.8	9.7	9.0	18.8	16.7	15.3	6.9
V_0	1.1	2.9	5.5	4.8	0.4	60.3	34.6	36.0	24.6
Ver	16.7	16.7	16.7	0.0	0.0	0.0	0.0	0.0	0.0
TOTAL (Moyenne)	5.7	8.2	7.6	9.9	7.7	33.4	25.4	27.3	12.9

Key Observations:

- **Qwen2.5's Breakthrough:** Fine-tuning more than tripled its score (9.9% → 33.4%). It learned to respect the strict single-lexeme constraints, excelling in POS-change functions.
- **Mistral's Overfitting:** Base model performed well (25.4%), but fine-tuning caused a severe performance drop (to 12.9%), indicating catastrophic forgetting of its initial linguistic priors.
- **FlauBERT's Stagnation:** Scores remained stable but low (5.7% → 8.2%), likely due to an insufficient learning rate causing a ceiling effect for meaningful parametric shifts.
- **Structure Matters:** Fine-tuning on unstructured Wikipedia data did not consistently beat base models, confirming that raw text exposure does not substitute for explicit structured annotations.
- **Universal Gaps:** Relations like *Ver, AntiVer,* and *Bon* proved universally difficult, suggesting severe data scarcity in models' pre-training corpora.

5. Conclusion

Mixed and Architecture-Dependent Results:

- **The Success:** Structural injection works: it more than tripled Qwen2.5's performance on understanding specialized lexical functions.
- **The Caveat:** Improving specific lexical competence does not smoothly translate into broader general NLU improvements (as seen on the COLE benchmark).

Future Work

Future research should explore alternative integration methods that are less prone to catastrophic forgetting: Soft-prompting, dynamically adapting LoRA hyperparameters, or integrating LF data directly during the standard alignment phases (RLHF/DPO) to balance specialized knowledge and robust general linguistic capabilities.

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